

ShanghaiTech University
School of Information Science and Technology

EE115A Lab Experiments

Experiment : Non - Ideal Op-Amps

2. Pre-Lab

To make the plots more readable and to save on printer toner, change the background color of your plots to white before printing your plot.

2.1 Slew Rate Measurement in Unity Gain Configuration

Slew Rate: _____

Attach your plot of V_{out} and V_{in} versus time to the end of the worksheet.

2.2 Gain and Bandwidth in Unity Gain Configuration

Gain A_0 : _____

Bandwidth f_{3dB} : _____

2.3 Gain and Bandwidth in Non-Inverting Amplifier Configuration

$R = 10\text{ k}\Omega$: Gain A_0 : _____

Bandwidth f_{3dB} : _____

$R = 100\text{ k}\Omega$: Gain A_0 : _____

Bandwidth f_{3dB} : _____

Attach your plot of the magnitude responses $20\log|V_{out}/V_{in}|$ of the unity gain amplifier from Problem 2.2 and the two non-inverting amplifiers to the end of the worksheet.

At approximately what frequency and gain do the three curves intersect on the plot?
What does this mean?